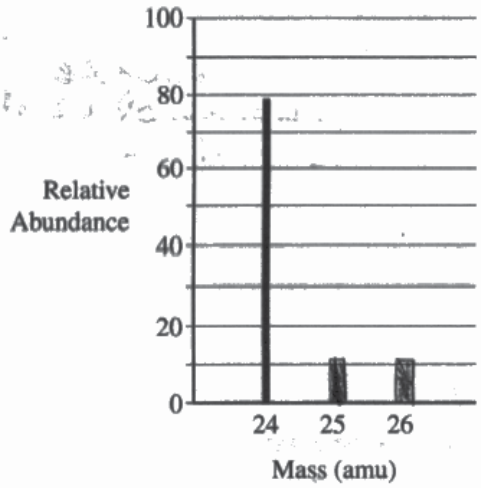
Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

Part A

## Question 1




i. 1 − .79 = .21 ÷ 2 = .105 ≈ 10.5%

ii. Each isotope has a different number of neutrons resulting in each one having a different mass.



Question 1 is continued on the next page.

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**Question 1**

**Part B**

i. $Na^+$ has a weaker attraction because it has a smaller charge, since Mg has a charge of +2 it is looking to gain 2 negative charges compared to Na only needing 1.

ii. $Na^+$ is bigger than $Mg^{+2}$ so the nucleus does not have as strong of a pull as Mg does.

**Part C**

$-\log(2.8\times10^{-4}) = pOH$

$pOH = 3.55$

$14 - 3.55 = 10.45$

$pH = 10.45$



Question 1 is continued on the next page.

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## Question 1

**Part D**

$1.85\times10^{-3}$ $Mg(NO_3)_2 + 2NaOH \longrightarrow 2NaNO_3 + MgOH$

.155 $6.475\times10^{-5}$

**Part E**

i. $K_{sp} = [Mg^{+2}][OH^-]^2$

ii. $Q = \frac{1.65\times10^{-4}}{6.475\times10^{-4}}$ $Q = .255$

iii. Yes a precipitate should form because $Q > K$ so the reverse reaction is favored and $Mg(OH)_2(s)$ will form.



Question 1 is continued on the next page.

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**Question 1**

**Part F**

The amount of undissolved $Mg(OH)_2$ should increase because the solution started as $Mg(NO_3)_2$ so adding $HNO_3$ would add a common ion and decrease the $K_{sp}$.



Continue to Question 2.